SALEH YAHYA

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EDUCATION

University of Illinois Urbana-Champaign Expected Dec 2027
M.S. in Computer Science
Relevant Courses: Parallel Programming, Distributed Systems, Scientific Visualization, Numerical Methods, Data Cleaning, etc

Lewis University Expected Dec 2026
B.S. in Computer Science
B.S. in Information Technology

PROFESSIONAL EXPERIENCE

Lewis University | *Teaching Assistant* *Aug 2025 – Dec 2025*
- Elected from 20+ students to assist for Object-Oriented Programming class, providing tutoring and coaching with Python & C++.
- Organized 2 weekly study sessions, for collaborative problem solving and stronger understanding of programming principles.
- Improved my students' performance, resulting in a 21% higher average grade compared to the previous semester.

Paidwork | *Machine Learning Engineer Intern* *Sep 2025 – Nov 2025*
- Improved biometric verification reliability by implementing liveness detection using camera sensors with Python & PyTorch.
- Increased document verification accuracy by refactoring object detection and optical character recognition models across multiple international identification formats, improving text extraction and layout detection by 8% using TensorFlow.
- Strengthened fraud prevention by developing anti spoofing computer vision perception models using depth estimation and texture based analysis, reducing fraudulent authentication from static image and replay attacks by 37% to a user base of 25M.

OveoAI | *Software Engineer Intern* *May 2025 – Aug 2025*
- Migrated and transformed data pipelines from MySQL to PostgreSQL using Python, increasing database performance by 18%.
- Tuned denial model that automatically classifies denial reasons and generates corrected resubmission suggestions.
- Improved claim denial model accuracy by 12% by incorporating payer rules and coding compliance checks.

RESEARCH

MPMC Queue Performance Based Benchmarking | *C++ 20, Python, SQL* | Publication | GitHub
- Benchmarked MPMC queue performance by comparing mutex synchronization against lock-free atomic implementations in C++.
- Designed and executed multi-threaded simulations across 200K–6M operations measuring throughput, latency benchmarking.
- Identified system bottlenecks including memory safety, performance degradation, and lower throughput as data and threads grew.
- Transformed data into database with Python & SQL, visualized data with dashboards, enhanced research efficiency by 20%.

PROJECTS

Lum – Financial Data Network & Analytics Platform | *Python,* Scikit-learn, *PostgreSQL, AWS, Redis, Nginx* | luminyy.com | GitHub
- Deployed fintech distributed system with AWS EC2, CloudWatch and CI/CD automation for production monitoring and scaling.
- Built asynchronous institutional ingestion workflows in Python, reducing financial management processing time by 40%.
- Reduced REST API response latency by 30% through Redis caching and optimized PostgreSQL indexing strategies.
- Developed ML financial analytics model using Scikit-learn for transaction classification & time series forecasting to 1k+ users.

Limit Orderbook Matching Engine | *C++, NoSQL, GTest* | GitHub
- Implemented a price, time priority limit order book in C++ processing microsecond level order matching for bid and ask levels.
- Designed custom priority queues and event driven logic supporting limit, market, and cancel orders enforcing FIFO execution.
- Simulated exchange style order flow and liquidity dynamics across thousands of order events under varying market conditions.

Raycast Inspired AI Overlay | *Python, Gemini API* | GitHub
- Built GUI AI overlay triggered by native hotkeys for instant LLM access in Python eliminating browser tab-switching disruptions.
- Designed multi-threaded architecture separating input listener, UI thread, and worker queue for thread safe execution.
- Configured HTTP streaming to reduce perceived latency by 40% and cut 3+ manual steps per query by streamlining workflows.

SKILLS

Languages: Python, C/C++, Java, JavaScript, SQL
Frameworks & Tools: Kafka, Redis, WebSocket, REST API, PyTest, GTest, Matplotlib
Infrastructure: AWS, UNIX/Linux, CUDA, PostgreSQL, NoSQL, GCC, Nginx, Jenkins, Docker, Git, GitHub Actions
Machine Learning: PyTorch, TensorFlow, Scikit-learn, Numpy, Pandas